ModelIt!

Student Activity Pack

Can an Algorithm Be Racist?

Modeling Bias in Artificial Intelligence and Machine Learning

Grade 11L1 | NGSS HS-ETS1-1, HS-ETS1-4 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Do you think a computer program can be biased or unfair? How is that possible if it's just following mathematical rules?**

2. **If an algorithm is trained on data from a world with racial discrimination, what do you think it will learn?**

3. **Should algorithms be used to make important decisions about people's lives — like who gets a job, a loan, or bail from jail? Why or why not?**

4. **What do you think the word 'fair' means when applied to an algorithm — equal treatment, equal outcomes, or something else?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Training Data Composition | Feature Selection | Algorithm Fairness Constraint | Prediction Accuracy | Disparate Impact Ratio

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Biased Historical Data

Training data: 20 years of historical hiring with demographic imbalance | No fairness constraints

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Debiased Training Data

Training data: Balanced demographic representation | No fairness constraints

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Fairness-Constrained Algorithm

Training data: Original biased data | Fairness constraint: Equal false positive rates

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Feedback Loop Amplification — The tendency for biased algorithmic decisions to generate new biased data that further trains the algorithm — for example, if a predictive policing algorithm sends more officers to Black neighborhoods, more arrests occur there, generating data that 'confirms' the algorithm's prediction

Transparency Level — The degree to which the algorithm's decision-making process is interpretable and auditable — 'black box' deep learning models make accurate predictions but cannot explain WHY, making bias detection extremely difficult

Stakeholder Impact — The severity of consequences for individuals affected by biased algorithmic decisions — being wrongly denied a job is harmful; being wrongly denied medical treatment or bail is potentially life-threatening

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If a computer algorithm trained on historical data makes decisions about who gets a loan, who gets hired, or who goes to jail — and the historical data reflects decades of racial discrimination — is the algorithm racist, or is it just doing math? Use evidence from your simulation to support your answers.

1 How does your model demonstrate that mathematical objectivity does not guarantee fairness?

2 Why does removing race from the algorithm's feature set fail to eliminate racial bias?

3 What does your model reveal about the relationship between overall accuracy and demographic fairness — are they always in tension?

4 Who should be responsible for ensuring algorithms are fair — the developers, the companies that deploy them, or the government?

5 What limitations does your model have in capturing the full complexity of algorithmic discrimination in the real world?

8 STEM Challenge: Design a Fair Hiring Algorithm Audit

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YOUR ENGINEERING MISSION

Design an audit protocol for a company's AI hiring system that identifies bias, quantifies disparate impact, and recommends evidence-based corrections to achieve equitable outcomes.

THE SCENARIO

A major tech company uses an AI system to screen 500,000 job applications per year. An internal review found that the system recommends candidates from predominantly white zip codes at 3x the rate of candidates from predominantly Black zip codes, even after controlling for qualifications. The company hired your algorithmic fairness team to audit the system, identify sources of bias, and recommend corrections that achieve a Disparate Impact Ratio above 0.8 without significantly reducing the quality of hires.

Engineering Design Thinking

1. WHAT DATA WILL YOU NEED TO ACCESS TO CONDUCT A THOROUGH BIAS AUDIT, AND WHAT METRICS WILL YOU USE TO MEASURE FAIRNESS?

2. HOW WILL YOU DISTINGUISH BETWEEN FEATURES THAT GENUINELY PREDICT JOB PERFORMANCE AND FEATURES THAT ARE PROXIES FOR DEMOGRAPHICS?

3. WHAT FAIRNESS CONSTRAINT WILL YOU RECOMMEND, AND HOW DOES IT AFFECT OUTCOMES FOR BOTH MAJORITY AND MINORITY APPLICANTS?

4. HOW WILL YOU ENSURE THE CORRECTED ALGORITHM DOESN'T DEVELOP NEW BIASES OVER TIME AS IT PROCESSES NEW DATA?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

AI Ethics Researchers study the social impact of artificial intelligence and develop frameworks for responsible AI deployment, earning $90,000–$180,000/year at tech companies, universities, and policy organizations. Algorithmic Auditors inspect automated decision systems for bias and fairness compliance, earning $85,000–$160,000/year. Data Scientists who specialize in fairness-aware machine learning earn $100,000–$200,000/year at leading tech companies.